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SUPERSEDING
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FEDERAL SPECIFICATION

LACQUER, FLUORESCENT TYPE, SYNTHETIC BASE, IN PRESSURIZED CONTAINER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a fluorescent type, synthetic base lacquer in pressurized containers for general purpose interior or exterior application.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-E-543 - Enamel, Interior Undercoat, Tints and White.
TT-N-97 - Naphtha; Aromatic.
UU-C-282 - Chipboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60 - Tape, Packaging, Waterproof.

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8010

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 1210 - Fineness of Dispersion of Pigments in Vehicle System.
D 1296 - Odor of Volatile Solvents and Diluents.
D 2369 - Volatile, Nonvolatile Content (Ordinary Laboratory Oven).
D 3335 - Determination of Low Concentration of Lead in Paints by Atomic Absorption Spectroscopy.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Chemical Specialties Manufacturers Association (CSMA) Publications:

CSMA Aerosol Guide: Methods for Pressure Measurement in Metal Aerosol Containers.

Revised Flammability Test Methods For Aerosol Products (Adopted September 12, 1963), Sixth Edition, March 1971.

(Application for copies should be addressed to Chemical Specialties Manufacturers Association, Inc., 1001 Connecticut Avenue, N.W., Washington, DC 20036).

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of good quality, entirely suitable for the purpose intended (see 6.1). The lacquer shall be free from materials which will be toxic under normal conditions of use. Polychlorinated biphenyls shall not be used in these products.

3.1.1 Coloring matter. Daylight fluorescent pigments shall be used to impart fluorescent properties to the lacquer.

3.1.2 Vehicle. The resin portion of the vehicle shall be a high grade acrylic or styrene-acrylic resin, suitable for the coloring matter used. The thinner of the lacquer shall be in accordance with industry's latest practices.

3.1.3 Stabilizer. The fluorescent colors may be stabilized by incorporation of ultraviolet absorbers.

3.1.4 Propellant. The propellant shall be a hydrocarbon or a blend of hydrocarbons and/or halogenated hydrocarbon(s) as required to assure that the lacquer conforms to the requirements as specified herein. Vinyl chloride shall not be used in these products.

3.2 Dispenser.

3.2.1 Container. The container shall be a commercial type metal pressure container, generally known as an aerosol container, of such construction as to assure acceptance of the finished package by common carriers operating in interstate commerce. The capacity of the container shall be sufficient to contain the specified net contents of the product with outage in accordance with generally recognized safe commercial practice. Each container shall be labeled as stated in 5.4.3.

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3.2.2 Dispensing valve. The valve shall have a spray head which can be removed without releasing pressure from the aerosol container. The valve shall have a metering orifice accessible to cleaning with a common pin after removal of the spray head. The removable spray head shall contain orifices of such dimensions as to produce spraying properties as required in 3.4.2. The lower end of the dip tube shall have a maximum clearance of 9.5mm (3/8 inch) from the bottom rim of the can after storage for one week in the product (which allows for swelling of the tube).

3.2.3 Agitator. Each dispenser of the fluorescent lacquer shall contain one spherical agitator either 9.5 mm (3/8 inch), 15.9 mm (5/8 inch), or 22.5 mm (3/4 inch) in diameter, or the equivalent mass in smaller spheres or irregularly-shaped materials, made of material chemically nonreactive to mild acids and alkalies.

3.2.4 Cover cap. The valve shall be protected from accidental functioning and damage by a cover cap which shall be removable and replaceable. The color of the cover cap shall be an approximate match to the color of the lacquer in the container.

3.3 Pressurized fluorescent lacquer.

3.3.1 Color. The fluorescent lacquer shall be furnished in the following colors in accordance with Fed. Std. No. 595. (For other colors, see 6.3).

Green - 38901
Orange - 38903
Red - 38905
Yellow - 38907

3.3.2 Net contents. Unless otherwise specified in the contract or purchase order (see 6.2), each can shall contain a minimum of 368, and a maximum of 454 (13-16 avoir. ounces) net weight of contents at the discretion of the supplier, and the unit shall be considered a nominal one-pint can. The contents shall meet the total solids specified in the appropriate column of Table II which requires that each can, regardless of net contents, shall have the same minimum amount of solids per can.

3.3.3 Pressure. When tested as specified in 4.4.3.9, the aerosol containers shall not burst or show evidence of leakage, distortion, or any other defects at an equilibrium pressure which shall not exceed 140 p.s.i.g. at 55 ± 1°C (130 ± 2°F). Any failure of this test shall be cause for rejection of the batch.

3.4 Qualitative requirements.

3.4.1 Condition in container. The lacquer in the pressurized container shall be easily redispersed by normal shaking. There shall be no clogging of the spray nozzle when tested in 4.3.3.1.

3.4.2 Spraying properties. When tested as in 4.3.3.2, the fluorescent lacquer shall spray evenly at temperatures from 16 to 32°C (60 to 90°F), and shall deliver a spray of normal-size particles. A minimum of 98 percent by weight of the content shall be delivered to a vertical surface. The fluorescent lacquer film shall dry to a smooth, even finish, free from orange peel, sags, specks, or any irregularities.

3.4.3 Adhesion (Tape test). When tested as in 4.4.3.5, the lacquer film shall adhere to the primed panel.

3.4.4 Odor. The odor of the lacquer, wet or dry, shall not be objectionable when tested in accordance with Table III.

3.4.5 Flexibility. The lacquer shall show no cracking, flaking, or loosening of the film from the panel when tested as in 4.4.3.4.

3.4.6 Self-lifting. When tested in accordance with Table III at 1, 4, 8, and 24 hours after recoating, there shall be no lifting of the film.

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3.4.7 Storage stability. At all times during the year following the date of acceptance, the aerosols shall be useable, as determined by:

(a) Condition in container (see 3.4.1).
(b) Spraying properties (see 3.4.2).
(c) Adhesion (see 3.4.3).
(d) Color matching (see 3.3.1).
(e) Drying time (see Table I).
(f) Pressure in container (see 3.3.3).
(g) Valve operation (see 3.2.2).
(h) There shall be no leaks, rust inside the container, nor other deficiencies which may impair useability.

3.5 Quantitative requirements. The lacquer shall conform to the quantitative requirements of table I and table II.

TABLE I. Quantitative requirements of the fluorescent lacquer

Characteristics	Requirements	
	Minimum	Maximum
Drying time:		
Dust free (minutes)	--	5
Tack free (minutes)	--	10
Dry hard (hours)	--	2
Fineness of grind	5	--
Fading, percent (southern exposure)		
Two-coat application 0.037mm (0.0015 inch thick), 3 months	--	50
Three or more coats 0.076mm (0.003 inch thick), 6 months	--	50
Lead, percent of non-volatile (as Pb metal)		0.5
Gloss		15

TABLE II. Physical properties

Net wt. Content per container	Nonvolatile percent by weight	Volatile percent by weight	Solids by weight per container minimum
368 g (13 oz)	15.0 minimum	85.0 maximum	55.3 g (1.95 oz)
397 g (14 oz)	13.9 minimum	85.1 maximum	55.3 g (1.95 oz)
426 g (15 oz)	13.0 minimum	87.0 maximum	55.3 g (1.95 oz)
454 g (16 oz)	12.2 minimum	87.8 maximum	55.3 g (1.95 oz)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling for visual examination and testing shall be in accordance with MIL-STD-105.

4.2.1 Visual examination. Inspection level S-4, AQL 2.5 defects per hundred units. Check for the following: Leaks, rust, dents, bruises, scratches, film defects, improper markings, ease of operation (shake and spray), and other deficiencies.

4.2.2 Tests. Inspection level S-3, AQL 2.5 defects per hundred units for each test performed on end items (filled and pressurized dispensers). Failure to pass a test shall constitute a defect.

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4.3 Inspection. Inspection shall be performed in accordance with method 1011 of Fed. Test Method Std. No. 141.

4.4 Test procedures.

4.4.1 Testing of the aerosol under this specification shall be for the purpose of acceptance of individual lots. Acceptance testing shall consist of all tests covered by this specification.

4.4.1.1 Supplier tests in process, in place of complete tests on samples of the finished dispensers, may be accepted where applicable for the purpose of lot acceptance, except spraying properties, color, and drying time must be tested after packaging.

4.4.2 Test concentrate. With the container well-agitated and in an upright position, pierce the top with a very fine hole (with a needle point), and allow the propellant to escape. After a few minutes, the hole can be enlarged, and as soon as there is no further evidence of gas escaping, the top of the container can be cut open. Allow to stand until the contents are at room temperature, then transfer the contents quantitatively into a tared beaker using several washes of naphtha conforming to TT-N-97. Evaporate on a water bath until no more propellant is liberated, and until the material weighs less than 40 percent of the original weight of the contents, including propellant. Then readjust to exactly the following percent of original weight with aromatic naphtha conforming to TT-N-97.

Nominal can contents	Percent
368 g (13 oz)	40.0
397 g (14 oz)	42.6
426 g (15 oz)	45.7
454 g (16 oz)	49.2

This compensates for the difference in solids required by table II for cans of different nominal net contents, and makes all test concentrates equivalent for further testing.

4.4.2.1 For the qualitative tests, the manufacturer may use the lacquer concentrate as it exists before packaging and pressurization, provided it is adjusted as specified in 4.3.2.

4.4.3 The tests shall be conducted as specified in table III. Unless otherwise specified, all tests shall be performed at a temperature of 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 5 percent.

TABLE III.

Characteristics	Requirement reference	Applicable test method: Fed. Test Method Std. No. 141	ASTM	Paragraph reference
Color	3.3.1	4250	----	4.4.3.7
Pressure	3.3.3	------	----	4.4.3.9
Condition in container	3.4.1	------	----	4.4.3.1
Spraying properties	3.4.2	------	----	4.4.3.2
Drying time	Table I	------	----	4.4.3.3
Flexibility	3.4.5	------	----	4.4.3.4
Adhesion	3.4.3	6304	----	4.4.3.5
Odor 1/	3.4.4	------	D 1296	-------
Fineness of grind 1/	Table I	------	D 1210	-------
Fading	Table I	------	----	4.4.3.6
Self-lifting	3.4.6	6252	----	-------
Volatile matter	Table II	------	D 2369	-------
Nonvolatile matter	Table II	------	D 2369	-------
Gloss	Table I	6101	----	4.4.3.7
Lead content	Table I	------	D 3335	-------

1/ Tests determined on concentrate (without propellant).

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4.4.3.1 Condition in container. Shake the container vigorously for 1 minute, hold it in a vertical position, and spray the lacquer on clean plate glass. Examine for compliance as specified in 3.4.1.

4.4.3.2 Spraying properties. Use a 10 by 30 cm (4 by 12-inch) steel panel, prepared in accordance with method 2011 of Fed. Test Method Std. No. 141. Weigh the full can before beginning the operation. Set the panel at an angle of approximately 15° from the vertical. Hold the dispenser in an upright position, parallel to the panel, at a distance of 10 to 15 cm (4 to 6 inches). With the valve fully open, move the dispenser horizontally from one side of the panel to the other. Continue to spray in overlapping horizontal strokes from the top to the bottom of the panel. Apply two thin coats (each coat approximately 13 um (0.0005 inch) thickness, one after the other. The second thin coat shall be applied by spraying vertically in overlapping strokes from side to side. After application of the second thin coat, place the panel in a vertical position, allow to dry for 5 minutes, then evaluate for compliance with 3.4.2. Continue to spray the contents until the pressure is exhausted. Weigh the can. Open the can by any suitable method and wash out the left-over lacquer with naphtha conforming to TT-N-97, then allow to dry for at least 2 hours. Weigh the can, then calculate the leftover as follows:

$$\text{Percent leftover} = \frac{B - C}{A - C} \times 100$$

Where: A = weight in grams of the full can.
B = weight in grams of empty can after spraying.
C = weight in grams of can thoroughly washed.

Evaluate for compliance with 3.4.2.

4.4.3.3 Drying time. Examine the drying time for the lacquer on panels as prepared in 4.4.3.2, and evaluate for compliance with the requirements specified in table I.

4.4.3.4 Flexibility. Spray a coat 76.2 um (0.003 inch) dry-film thickness consisting of three or more light passes on a steel panel prepared in accordance with method 2011, allowing 2 or 3 minutes drying time at 21°C (70°F) between passes. After the third pass, allow to dry for 2 hours, and then bake for 1 hour at 82°C (180°F). Cool the panel at room temperature for 1/2 hour, bend over a 6.35 mm (1/4 inch) mandrel, and examine for compliance with 3.4.5.

4.4.3.5 Adhesion (tape test). Prepare a steel panel in accordance with method 2011 of Fed. Test Method Std. No. 141. Brush a coat of enamel undercoat conforming to TT-E-543 to an 25 um (.001 inch) dry film thickness. Spray two thin coats of lacquer as described in 4.4.3.2. Allow the last coat to dry for 48 hours, then score a line through to the enamel undercoat across the width of the panel using a sharp knife. Tape the film, perpendicular to and across the score line with waterproof, pressure sensitive tape conforming to PPP-T-60, Type IV. Press the tape in firm contact using thumb pressure. Grasp the free end of the tape and at rapid speed strip it from the film by pulling back from the panel at approximately 180° angle. Examine the second film and evaluate for compliance with 3.4.3.

4.4.3.6 Light fastness. A warranty from the supplier is required, that the fluorescent film, exposed as specified in table I, shall not show a reduction greater than 50 percent in luminance (brightness) following exposure, as compared against an unexposed panel.

4.4.3.7 Color. Spray a coat of 76 um (0.003 inch) dry-film thickness on a clean white carrara glass plate or on white, sealed paper stock. This can generally be accomplished by three or more light passes, as described in 4.4.3.2. Measure the color and gloss as specified in Table III, and evaluate for compliance with 3.3.1 and Table I.

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4.4.3.8 Dispenser. The dispenser and its component parts shall be inspected and evaluated for compliance with 3.2.

4.4.3.9 Pressure. Containers filled for shipment shall be submerged in a water bath maintained at 55 ± 1°C (130 ± 2°F). The pressure shall be measured after 15 minutes of immersion in accordance with the method described in the CSMA Aerosol Guide, "Methods for Pressure Measurement in Metal Aerosol Containers," method 2. Evaluate for compliance with 3.3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging. (see 3.2).

5.2 Packing.

5.2.1 Level A. Twelve dispensers of the same description shall be packed in a close-fitting box conforming to PPP-B-636, V3c; or PPP-B-640, class 2. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of the V3c box shall not exceed the weight limitations of PPP-B-636.

5.2.2 Level B. Twelve dispensers of the same description shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Separators the full height of the box shall be required conforming to UU-C-282, 0.040 in. minimum thickness. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C. The lacquer packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to destination. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. Packages and shipping containers shall be marked in accordance with Fed. STd. No. 123.

5.3.2 Military activities. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special marking. Legible commercial marking shall be acceptable. Cans overcoated to obliterate original marking shall not be acceptable. Marking shall be lithographed on the unit container or printed on a label securely affixed to the container and shall include the following:

(a) Color number of Fed. Std. No. 595.
(b) Net weight in ounces.
(c) Name of material.
(d) Directions for use, including surface preparation.
(e) A formula in general terms.
(f) "For best results, use at temperatures between 16 and 32°C (60 and 90°F)".
(g) Date of manufacture.
(h) "Do not puncture or incinerate".
(i) "Keep from fire or flame".
(j) "Use with adequate ventilation".
(k) "Avoid repeated contact with skin".
(l) "Store below 49"C (120°F)".
(m) "Keep from children".

6. NOTES

6.1 Intended use. Fluorescent lacquers in pressurized containers provide a quick, convenient method for painting small object for bright daytime visibility. They may also be used in areas where night visibility is required. In order to be visible at night, a source of light must be present. They can be applied on metal, wood, paper, linoleum, plaster, masonry, stone, glass, leather, fiber, or previously-painted surfaces. For maximum corrosion resistance on metals, a quick-drying lacquer-resistant inhibiting primer should be applied as a first coat. Surfaces to be sprayed should be clean, dry, and free from oily or greasy films. High-gloss surfaces should be scuffed, with steel wool before spraying. The fluorescent material shall always be applied over a white undercoat or surface.

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6.1.1 If additional durability is desired, apply a clear acrylic or modified acrylic film containing an ultraviolet absorber over the fluorescent coating. Reduced fluorescence and visibility is then to be expected.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color required (see 3.3.1 and 6.3).
(c) Levels of packaging and packing (see 5.1 and 5.2).

6.3 Color not covered by Fed. Std. No. 595. The supplier shall furnish a color chip matching the color agreed upon between the purchaser and supplier. This chip may not meet the gloss requirement of Table I.

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
COM-NBS
VA-DMS
INT-AFS

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